

Illumination-Conditioned Image Restoration for Low-Light Industrial Anomaly Inspection

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Abstract—Low illumination degrades both the visibility of industrial images and the reliability of downstream anomaly inspection. We introduce ICRA-MIRNet, an illumination-conditioned residual adapter around a pretrained MIRNet-v2 restoration backbone. The gate spatially modulates the pretrained restoration residual, and we jointly fine-tune the backbone and gate using only normal industrial training images under a deterministic camera-inspired low-light degradation protocol. Across three runs on 15 MVTec AD categories, ICRA-MIRNet obtains 28.472 ± 0.243 dB PSNR and 0.8946 ± 0.0015 SSIM, while raising class-macro anomaly AUROC from 0.6712 ± 0.0050 on dark inputs to 0.7378 ± 0.0066 after restoration. This benefit is not universal: restoration changes AUROC from 0.7400 to 0.7119 under mild degradation, from 0.6326 to 0.6584 under severe degradation, and from 0.6435 to 0.4777 on six-category MPDD without adaptation. These results show that improved visibility does not guarantee improved anomaly discrimination and explicitly separate controlled synthetic evidence from real low-light deployment claims.

Index Terms—Low-light enhancement, industrial inspection, anomaly detection, image restoration, domain robustness.

I. Introduction

Industrial anomaly inspection is commonly developed from well-exposed training images, whereas deployed cameras may experience illumination gradients, photon noise, and read noise. Image enhancement can improve visibility, but a restoration model that changes defect evidence may increase rather than reduce inspection risk. Consequently, restoration fidelity and downstream anomaly discrimination must be evaluated together.

MIRNet-v2 preserves high-resolution representations while exchanging contextual information across multiple resolutions and supports several restoration tasks, including low-light enhancement [1]. However, its generic enhancement output is not conditioned on the spatial illumination pattern of a specific industrial image. We therefore wrap its output residual with a learnable illumination-conditioned gate. This design retains the pretrained mapping when the gate approaches one and permits spatially selective correction after low-memory adaptation.

The experiments use MVTec AD [2] as the primary industrial benchmark and MPDD [3] for cross-dataset evaluation. The low-light observations are generated deterministically from the original clear images. This protocol supports paired fidelity measurements, but it does not constitute evidence on real night-time factory imagery.

The intended contributions are:

- an illumination-conditioned residual formulation that adapts a pretrained MIRNet-v2 model on a single 8-GB GPU;

- a leakage-controlled protocol that trains only on normal images and jointly reports PSNR, SSIM, and class-macro anomaly AUROC;
- three-seed, component-ablation, cross-degradation, and MPDD cross-dataset evaluations with resumable configurations and auditable progress logs.

II. Related Work

A. Low-Light Image Enhancement

The LOL dataset introduced paired low- and normal-light images for supervised enhancement [4]. MIRNet-v2 subsequently provided a multiscale architecture for restoration and enhancement with high-resolution feature preservation [1]. Our work does not propose a replacement restoration backbone; it studies whether a lightweight illumination-conditioned residual adapter can preserve industrial defect evidence under a fixed low-light protocol.

B. Industrial Anomaly Inspection

MVTec AD contains normal training images and normal/anomalous test images across object and texture categories [2]. MPDD targets painted metal parts acquired under more complex manufacturing conditions [3]. We use image-level anomaly discrimination as a downstream probe. The probe is deliberately fixed across clear, dark, and restored inputs so that only the image domain changes.

III. Method

A. Deterministic Low-Light Formation

Let $\mathbf{y} \in [0, 1]^{3 \times H \times W}$ denote a clear industrial image. A seeded degradation operator produces

$$\mathbf{x} = \text{clip}(\text{Poisson}(255 \mathbf{y}^\gamma \odot \mathbf{m}) / 255 + \epsilon, 0, 1), \quad (1)$$

where γ is sampled from a fixed interval, $\mathbf{m}$ is a smooth spatial illumination plane, and ϵ is zero-mean Gaussian read noise. For the training-range protocol, $\gamma \sim \mathcal{U}(2, 4)$. The illumination plane uses horizontal and vertical slopes sampled independently from $\mathcal{U}(-0.30, 0.30)$ and an offset from $\mathcal{U}(0.55, 0.85)$, followed by clipping to $[0.35, 1]$. Poisson noise is sampled at a 255-count scale, and the read-noise standard deviation is sampled from $\mathcal{U}(0, 0.02)$. The degradation seed is the global seed plus the image index, so each indexed pair is reproducible.

B. Illumination-Conditioned Residual Adapter

Let $B(\mathbf{x})$ be the pretrained MIRNet-v2 output and

$$\ell(\mathbf{x}) = 0.299\mathbf{x}_R + 0.587\mathbf{x}_G + 0.114\mathbf{x}_B \quad (2)$$

be luminance. A two-layer convolutional network predicts a spatial gate

$$\mathbf{g} = \sigma(G(1 - \ell(\mathbf{x}))). \quad (3)$$

The final restoration is

$$\hat{\mathbf{y}} = \mathbf{x} + \mathbf{g} \odot (B(\mathbf{x}) - \mathbf{x}). \quad (4)$$

The final gate layer is initialized with zero weights and bias eight, so the initial gate is close to one and Eq. (4) initially approximates the pretrained backbone.

C. Optimization

The training loss is

$$\mathcal{L} = \|\hat{\mathbf{y}} - \mathbf{y}\|_1 + \lambda_{\nabla} \|\nabla \hat{\mathbf{y}} - \nabla \mathbf{y}\|_1 + \lambda_g \|\mathbf{g} - (1 - \ell(\mathbf{y}))\|_1. \quad (5)$$

The default audited configuration uses $\lambda_{\nabla} = 0.2$ and $\lambda_g = 0.05$.

D. Fixed Downstream Anomaly Probe

For each category c , a frozen ImageNet-pretrained ResNet-18 [5] extracts global features ϕ . The official torchvision resize, center-crop, and ImageNet normalization are applied. A diagonal Gaussian is fitted only to clear normal training features:

$$s_c(\mathbf{z}) = \frac{1}{d} \sum_{j=1}^d \frac{(\phi_j(\mathbf{z}) - \mu_{c,j})^2}{\max(\sigma_{c,j}^2, 10^{-6})}. \quad (6)$$

The same probe scores clear, synthetically darkened, and restored test images. AUROC is calculated within each category and macro-averaged across categories.

IV. Experiments

A. Datasets and Protocol

The local MVTEC AD copy contains 3,629 normal training images and 1,725 test images across 15 categories. MPDD contributes 888 normal training images and 458 test images across six metal-part categories. The restoration model sees only the training/normal split. Test images never participate in restoration optimization or Gaussian fitting.

The main model processes 12,000 micro-batches using 128×128 crops, batch size two, four-micro-batch gradient accumulation, automatic mixed precision, and seeds 20260830–20260832. Thus, each uninterrupted main run performs 3,000 Adam parameter updates at a learning rate of 2×10^{-5} while jointly fine-tuning the backbone and gate. Each indexed training image receives a deterministic crop and degradation under the global seed. Full-resolution restoration uses overlapping 128×128 tiles with 16-pixel overlap. Training state is saved every 500 micro-batches and progress is logged every 100 micro-batches.

TABLE I

MVTEC AD results. ICRA-MIRNet reports mean $\pm$ sample standard deviation over seeds 20260830–20260832; the LOL-pretrained baseline and component ablations use seed 20260830. Values are inserted only from validated metrics.json files.

Method	PSNR $\uparrow$	SSIM $\uparrow$	Dark AUROC $\uparrow$	Restored AUROC $\uparrow$	Time $\downarrow$	Memory $\downarrow$
LOL-pretrained MIRNet-v2	18.168	0.6164	0.6674	0.7043	3.775 s	350.548 MiB
Fixed-gate fine-tuning	28.581	0.8960	0.6674	0.7472	4.062 s	386.249 MiB
ICRA-MIRNet without $\mathcal{L}_g$	28.744	0.8906	0.6674	0.7499	3.850 s	386.249 MiB
ICRA-MIRNet	28.472 $\pm$ 0.243	0.8946 $\pm$ 0.0015	0.6712 $\pm$ 0.0050	0.7378 $\pm$ 0.0066	3.034 $\pm$ 0.886 s	386.249 $\pm$ 0.000 MiB

TABLE II

Robustness and cross-dataset evaluation.

Evaluation domain	PSNR $\uparrow$	Restored AUROC $\uparrow$
MVTEC, mild degradation	21.514	0.7119
MVTEC, training-range degradation	28.472 $\pm$ 0.243	0.7378 $\pm$ 0.0066
MVTEC, severe degradation	21.466	0.6584
MPDD, training-range degradation	19.337	0.4777

B. Metrics

Restoration is measured by image-pooled PSNR and SSIM [6]. Inspection is measured by image-level AUROC within each category and then macro-averaged without category-size weighting for clear, dark, and restored inputs. Runtime is the image-pooled restoration latency. Peak allocated CUDA memory is reset and recorded for the evaluation phase only; it is not total process memory or peak training memory. Three-seed main-model results use the arithmetic mean and sample standard deviation.

C. Baselines and Ablations

The audited matrix contains: (i) the official LOL-pretrained MIRNet-v2 without industrial adaptation; (ii) backbone fine-tuning with a fixed gate $g = 1$; (iii) ICRA-MIRNet without the gradient loss; and (iv) the complete ICRA-MIRNet. Mild and severe degradation intervals test sensitivity outside the training range. A MVTEC-trained checkpoint is evaluated on MPDD without further adaptation.

V. Reproducibility and Evidence Boundaries

The cached outputs from official MIRNet-v2 inference on the 15-image LOL test set were independently re-evaluated with the authors' Python PSNR definition, yielding 24.7410 dB (24.74 dB when reported to two decimals). The industrial runner records configuration, checkpoints, training progress, per-image evaluation progress, per-category metrics, and final JSON output. A separate validator rejects missing categories and non-finite metrics before producing mean $\pm$ standard-deviation tables.

All industrial low-light inputs in this study are synthetic. Neither MVTEC AD nor MPDD is a paired night-time industrial dataset. Accordingly, the experiments can establish controlled degradation robustness and cross-dataset transfer, but they cannot establish performance in a real dark factory. A real paired or exposure-bracketed industrial test set remains necessary before any deployment claim.

VI. Conclusion

We presented an illumination-conditioned residual adapter and an audited protocol for jointly evaluating low-light restoration and industrial anomaly discrimination. Three MVTec AD runs show a mean AUROC gain of 0.0666 over training-range dark inputs. Severe degradation retains a smaller gain of 0.0258, whereas mild degradation decreases AUROC by 0.0281 and zero-adaptation transfer to MPDD decreases it by 0.1658. The mixed direction across degradation intensity and dataset shift cautions against treating restoration fidelity as a proxy for inspection reliability. The present evidence boundary is controlled synthetic low light; future work must add real exposure variation and an operational defect detector.

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